

EXAMPLE 15: Balance the equation:



H and Cl are not balanced, hence:



EXERCISE 31 Balance the following equations.

1. $\text{Na} + \text{Cl}_2 \longrightarrow \text{NaCl}$
2. $\text{CO} + \text{O}_2 \longrightarrow \text{CO}_2$
3. $\text{Al} + \text{O}_2 \longrightarrow \text{Al}_2\text{O}_3$
4. $\text{HgO} \longrightarrow \text{Hg} + \text{O}_2$
5. $\text{Mg} + \text{N}_2 \longrightarrow \text{Mg}_3\text{N}_2$
6. $\text{P}_4 + \text{O}_2 \longrightarrow \text{P}_4\text{O}_{10}$
7. $\text{Al} + \text{F}_2 \longrightarrow \text{AlF}_3$
8. $\text{PbO} + \text{C} \longrightarrow \text{Pb} + \text{CO}_2$
9. $\text{CH}_4 + \text{O}_2 \longrightarrow \text{CO}_2 + \text{H}_2\text{O}$
10. $\text{NaHCO}_3 \longrightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$
11. $\text{Ca} + \text{H}_2\text{O} \longrightarrow \text{Ca(OH)}_2 + \text{H}_2$
12. $\text{Fe} + \text{HCl} \longrightarrow \text{FeCl}_2 + \text{H}_2$
13. $\text{MgCO}_3 + \text{HCl} \longrightarrow \text{MgCl}_2 + \text{H}_2\text{O} + \text{CO}_2$
14. $\text{NaOH} + \text{H}_2\text{SO}_4 \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$
15. $\text{Mg} + \text{AgNO}_3 \longrightarrow \text{Mg(NO}_3)_2 + \text{Ag}$

EXERCISE 32 Write names for the elements and compounds in the equations for Exercise 31.

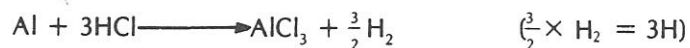
EXAMPLE 16: Balance the equation:



Both the H and Cl are not balanced. Balance the Cl first since H_2 contains no other elements, enabling H to be balanced last.



Balancing the H



The above equation is satisfactory but the fraction can be eliminated by multiplying by 2



EXAMPLE 17: Balance the equation:



NH_4^+ and NO_3^- ions appear on both sides of the equation. It is easier to balance the equation by balancing these groups, rather than dividing them into the elements of which they are made.

Neglecting these groups, C and O are balanced, but not H.

Balance the H in HNO_3 .



Simultaneously balance NH_4^+ and NO_3^-



EXERCISE 33 Balance the following equations.

1. $\text{H}_2\text{S} + \text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{SO}_2$
2. $\text{Al} + \text{AgNO}_3 \longrightarrow \text{Al(NO}_3)_3 + \text{Ag}$
3. $\text{NaOH} + \text{H}_3\text{PO}_4 \longrightarrow \text{Na}_3\text{PO}_4 + \text{H}_2\text{O}$
4. $\text{KClO}_3 \longrightarrow \text{KCl} + \text{O}_2$
5. $\text{NaOH} + \text{Fe(NO}_3)_2 \longrightarrow \text{Fe(OH)}_2 + \text{NaNO}_3$
6. $\text{Fe}_2\text{O}_3 + \text{CO} \longrightarrow \text{Fe} + \text{CO}_2$
7. $\text{Cu} + \text{HNO}_3 \longrightarrow \text{Cu(NO}_3)_2 + \text{H}_2\text{O} + \text{NO}_2$
8. $\text{NH}_3 + \text{CuO} \longrightarrow \text{Cu} + \text{H}_2\text{O} + \text{N}_2$
9. $\text{Al}_2(\text{CO}_3)_3 + \text{HCl} \longrightarrow \text{AlCl}_3 + \text{H}_2\text{O} + \text{CO}_2$
10. $\text{FeS} + \text{O}_2 \longrightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2$

EXERCISE 34 Write names for the elements and compounds in the equations for Exercise 33.

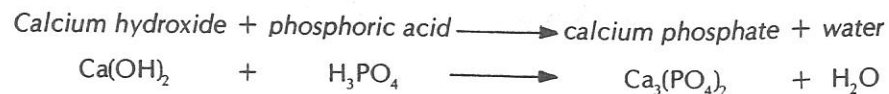
EXAMPLE 18: Balance the equation:



H and Cl are unbalanced, hence:



EXAMPLE 19: Balance the equation:



Balance Ca



Balance PO_4^{3-}



Balance H and O (other than that in PO_4^{3-})

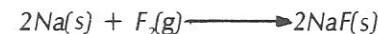


EXERCISE 35 Write fully balanced equations for the following.

1. hydrogen + oxygen $\longrightarrow$ water
2. carbon + oxygen $\longrightarrow$ carbon dioxide
3. sodium + chlorine $\longrightarrow$ sodium chloride
4. potassium + oxygen $\longrightarrow$ potassium oxide
5. magnesium + oxygen $\longrightarrow$ magnesium oxide
6. magnesium + hydrochloric acid $\longrightarrow$ magnesium chloride + hydrogen
7. ammonia + water $\longrightarrow$ ammonium hydroxide
8. zinc + hydrochloric acid $\longrightarrow$ zinc chloride + hydrogen
9. sodium + sulfuric acid $\longrightarrow$ sodium sulfate + hydrogen
10. calcium carbonate + hydrochloric acid $\longrightarrow$ calcium chloride + water + carbon dioxide
11. hydrogen + nitrogen $\longrightarrow$ ammonia
12. aluminium + oxygen $\longrightarrow$ aluminium oxide
13. calcium oxide + water $\longrightarrow$ calcium hydroxide
14. calcium hydroxide + carbon dioxide $\longrightarrow$ calcium carbonate + water
15. ammonia + sulfuric acid $\longrightarrow$ ammonium sulfate
16. silver nitrate + magnesium chloride $\longrightarrow$ silver chloride + magnesium nitrate
17. potassium nitrate $\longrightarrow$ potassium nitrite + oxygen
18. sodium hydroxide + carbon dioxide $\longrightarrow$ sodium carbonate + water
19. potassium bicarbonate $\longrightarrow$ potassium carbonate + water + carbon dioxide
20. zinc + silver nitrate $\longrightarrow$ zinc nitrate + silver

EQUATIONS SHOW THE RATIO IN WHICH MOLECULES, ATOMS OR IONS REACT.

EXAMPLE 20: What information is given by the equation below?



For every two atoms of sodium in a metallic lattice that react,

- one molecule of fluorine reacts
- two sodium ions and two fluoride ions in an ionic lattice are formed.

EXERCISE 36 What information is given by each of the equations below?

1. $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{H}_2\text{O}(\text{g})$
2. $\text{C(s)} + \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g})$
3. $2\text{KHCO}_3(\text{s}) \longrightarrow \text{K}_2\text{CO}_3(\text{s}) + \text{H}_2\text{O}(\text{g}) + \text{CO}_2(\text{g})$
4. $\text{Zn(s)} + 2\text{AgNO}_3(\text{aq}) \longrightarrow \text{Zn(NO}_3)_2(\text{aq}) + 2\text{Ag(s)}$
5. $3\text{H}_2(\text{g}) + \text{N}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$

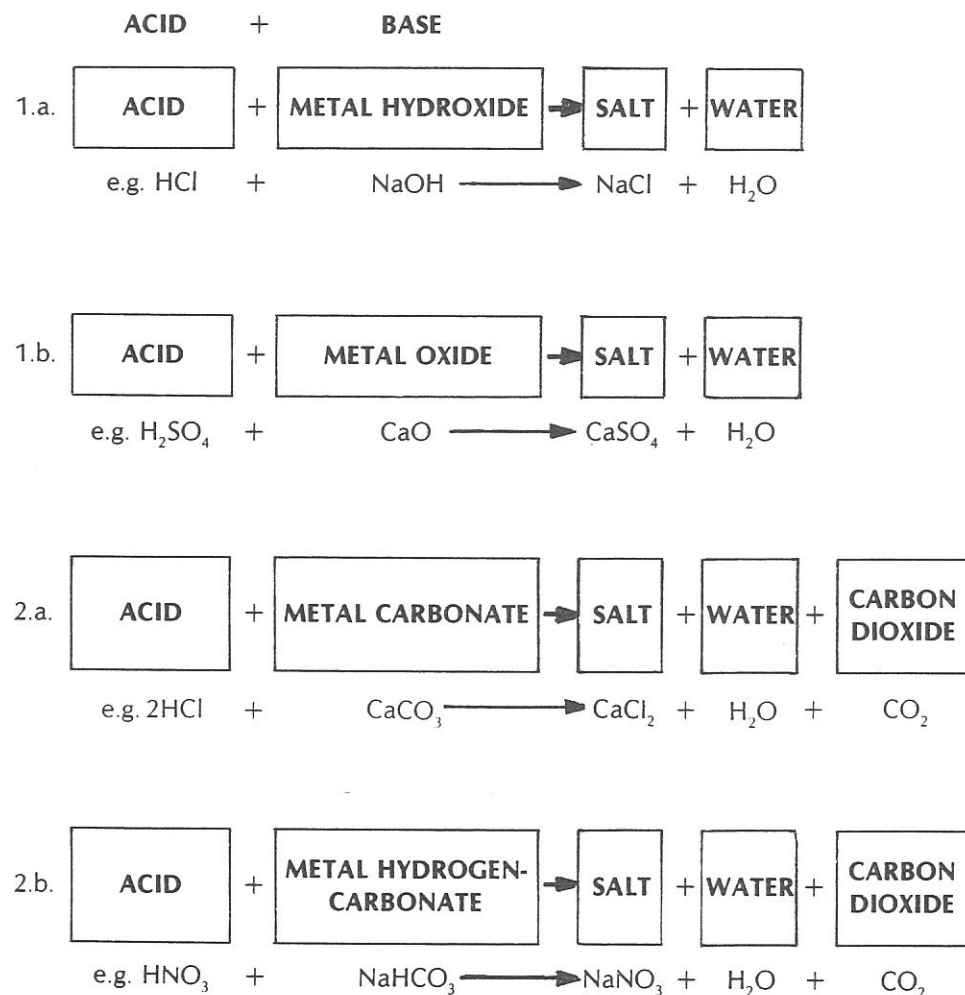
CHAPTER 7: General Reactions

Substances with similar chemical properties will react in similar ways and therefore many chemical reactions can be classified into general reaction types. Knowledge of general reactions enables products of specific reactions to be predicted.

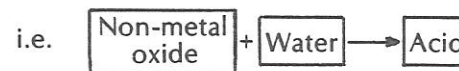
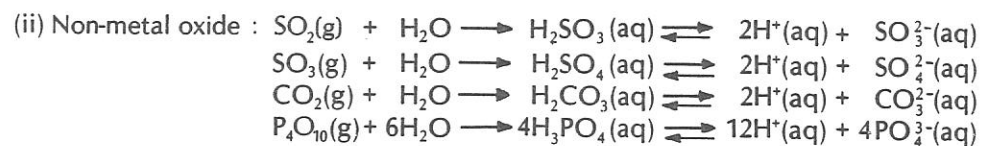
The most common general reaction types will be treated here.

NEUTRALIZATION

Neutralization is the term used to describe the reaction between an *acid* and a *base*. The general equations for the many different forms of acids and bases are indicated here:



An **ACID** is a substance that produces hydrogen ions (H^+) in solution.



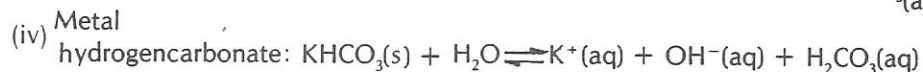
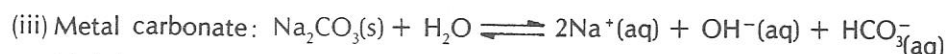
- NOTE:** 1. The use of $\rightleftharpoons$ indicates that the reaction has not gone to completion.
 2. The acid in any of the general neutralization reactions above could be a non-metal oxide, in which case the salt formed is due to the acid produced when the non-metal oxide dissolves in water.
 e.g. $\text{SO}_2 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$ (equation type (1.a.))
 3. Sulfur dioxide is a major pollutant causing "acid rain".

The following acids should be memorised:

NAME OF ACID	FORMULA	USE
Hydrochloric acid	HCl	Cleans bricks and iron
Sulfuric acid	H_2SO_4	Car batteries, make fertilizers
Nitric acid	HNO_3	Make fertilizers, explosives
Phosphoric acid	H_3PO_4	Make fertilizers
Acetic acid	CH_3COOH	Vinegar
Carbonic acid	H_2CO_3	—
Sulfurous acid	H_2SO_3	—

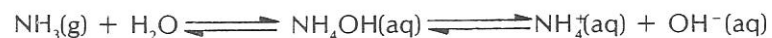
Acids taste sour, feel rough and are corrosive.

A **BASE (or alkali)** is a substance that produces hydroxide ions (OH^-) in solution.



NOTE:1. Although they contain no metal ions, both ammonium carbonate, $(\text{NH}_4)_2\text{CO}_3$, and ammonium hydrogencarbonate, NH_4HCO_3 , also behave as bases.

2. Ammonia solution is a common base. Its property of being a base can be seen in the following:



The following bases should be memorised:

COMMON NAME OF BASE	FORMULA	USE
Caustic soda	NaOH	Make soap, clean drains
Milk of magnesia	$\text{Mg}(\text{OH})_2$	Relieve an upset stomach
Ammonia	NH_3	Make fertilisers, explosives
Washing soda	Na_2CO_3	Make glass, soften water
Baking soda	NaHCO_3	Bake cakes
Quicklime	CaO	} Make lime mortar, plaster
Slaked lime	$\text{Ca}(\text{OH})_2$	

Bases taste bitter, feel slippery and are corrosive. They are used in household cleaners, dishwasher powder and oven cleaners to remove grease and fats.

EXERCISE 37: ACIDS AND BASES

Write the formula or name of each of the following substances and then classify it as an acid or a base:

- | | | |
|--------------------------------|----------------------------|----------------------|
| 1. Carbon dioxide | 5. Barium carbonate | 9. NH_3 |
| 2. Magnesium oxide | 6. KOH | 10. Nitrogen dioxide |
| 3. Li_2O | 7. H_3PO_4 | |
| 4. $\text{Ca}(\text{HCO}_3)_2$ | 8. Sulfur trioxide | |

A **SALT** is produced by the combination of the positive ion from the base and the negative ion from the acid.

For example, $\text{NaCl}(\text{aq})$ is the salt formed when $\text{HCl}(\text{aq})$ is reacted with $\text{NaOH}(\text{aq})$.

WATER is produced in neutralization through the combination of the hydrogen ion, $\text{H}^+(\text{aq})$, from the acid and the hydroxide ion, $\text{OH}^-(\text{aq})$, from the base.

EXAMPLE 21: Write the balanced equation for the reaction between dilute hydrochloric acid and copper(II) oxide solid.

ANSWER: By equation type (1.b.)



EXERCISE 38: NEUTRALIZATION

Write the balanced equations for the reactions produced when the following substances are mixed:

- Dilute sulfuric acid + potassium hydroxide solution.
- Dilute hydrochloric acid + calcium hydroxide.
- Dilute hydrochloric acid + magnesium oxide.
- Sulfur dioxide gas + sodium hydroxide solution.
- Calcium carbonate + dilute nitric acid.
- Acetic acid solution + iron(II) oxide.
- Aluminium hydrogencarbonate + sulfuric acid solution.
- Aluminium oxide + dilute hydrochloric acid.
- Calcium hydroxide + phosphoric acid solution.
- Potassium hydroxide solution + sulfur trioxide gas.
- Sulfurous acid solution + iron(III) hydrogencarbonate solution.
- Barium hydroxide solution + phosphoric acid solution.
- Nickel carbonate + dilute sulfuric acid.
- Tin(II) hydrogencarbonate + dilute hydrochloric acid.
- Dilute acetic acid + zinc hydroxide.
- Ammonia solution + dilute nitric acid.
- Carbonic acid + sodium hydroxide solution.
- Tetraphosphorus decaoxide + potassium hydroxide solution.
- Ammonium hydrogencarbonate solution + dilute hydrochloric acid.
- Dilute nitric acid + iron(III) hydroxide.

INDICATORS

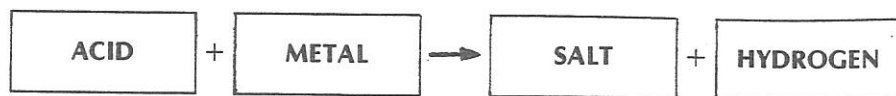
An **INDICATOR** is a substance, often a plant dye, which has a characteristic but different colour in acidic and basic solutions.

INDICATOR	COLOUR IN ACID	COLOUR IN BASE
Litmus	Red	Blue
Methyl orange	Red	Yellow
Phenolphthalein	Colourless	Purple

EXERCISE 39: INDICATORS

State the colour of litmus, methyl orange and phenolphthalein in solutions of each of the substances listed in Exercise 37.

REACTION BETWEEN AN ACID AND A METAL



In this reaction the metal is displacing the hydrogen from the acid and therefore can be considered as a special example of displacement reactions (see p. 39). Therefore **ONLY METALS ABOVE HYDROGEN** on the Activity Series will obey this reaction (see inside back cover).

NOTE: 1. Neither nitric acid nor concentrated sulfuric acid obey this equation.
2. When Fe reacts by this equation it produces Fe^{2+} ions.
3. When Sn reacts by this equation it produces Sn^{2+} ions.
4. When Pb reacts by this equation it produces Pb^{2+} ions.

EXERCISE 40: ACID-METAL REACTIONS

Write balanced equations for the reactions produced when the following substances are mixed. If a reaction does not occur, indicate by writing "no reaction".

1. Zinc + dilute hydrochloric acid.
2. Magnesium + dilute sulfuric acid solution.
3. Aluminium + hydrochloric acid.
4. Phosphoric acid + calcium.
5. Copper + dilute hydrochloric acid.
6. Iron + dilute sulfuric acid.
7. Aluminium + acetic acid solution.
8. Dilute sulfuric acid + silver.
9. Dilute sulfuric acid + lead.
10. Nickel + dilute hydrochloric acid.

REACTION BETWEEN A REACTIVE METAL AND WATER



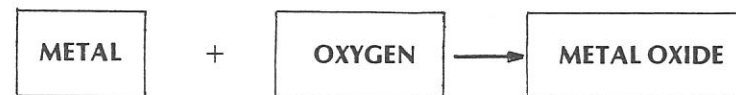
This reaction occurs *only for the elements at the top* of the Activity Series (*potassium, sodium and calcium*). However, if the water is converted to steam, then other metals (e.g. Mg, Al, Fe, Zn), towards the top of the Activity Series (see inside back cover) can be made to react in this way to form a metal oxide and hydrogen.

EXERCISE 41: WATER-METAL REACTIONS

Write the balanced equation for the reaction produced when the following substances are mixed:

1. Sodium + water.
2. Calcium + water.

REACTION BETWEEN A METAL AND OXYGEN



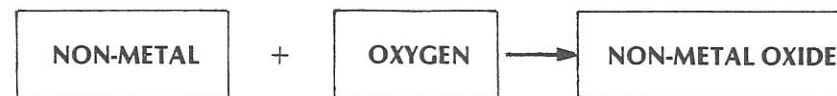
This reaction is often referred to as **OXIDATION**, although if it proceeds very rapidly it is said to be **COMBUSTION** or *burning*. The ability to react with oxygen will vary according to the Activity Series (see inside back cover).

EXERCISE 42: OXYGEN-METAL REACTIONS

Write the balanced equations for the following reactions.

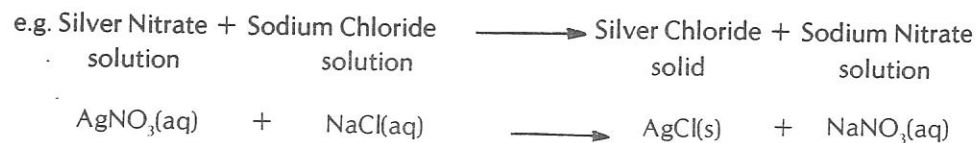
1. Reaction of calcium and oxygen.
2. Oxidation of zinc by oxygen.
3. Burning of magnesium.
4. Production of iron(III) oxide from the oxidation of iron.
5. Production of lead(II) oxide from the oxidation of lead.
6. Formation of copper(II) oxide from the oxidation of copper.
7. Oxidation of nickel by oxygen.
8. Production of iron(II) oxide from the oxidation of iron.
9. Formation of mercury(II) oxide from the oxidation of mercury.
10. Oxidation of tin to tin(II) oxide by oxygen.

REACTION BETWEEN A NON-METAL AND OXYGEN



PRECIPITATION REACTIONS

When two different salt solutions are mixed, a precipitate will occur if, by combination of a positive ion and a negative ion, a substance of low solubility is produced.



To predict products of precipitation reactions a knowledge of *solubility rules* is required. The more common rules are indicated here.

SOLUBILITY OF IONIC COMPOUNDS IN WATER

Soluble Compounds	Exceptions
Halides $\left[\begin{array}{l} \text{Chlorides (Cl}^-) \\ \text{Bromides (Br}^-) \\ \text{Iodides (I}^-) \end{array} \right]$	$\text{Ag}^+, \text{Pb}^{2+}, \text{Hg}_2^{2+}, \text{Cu}^+$
Nitrates (NO_3^-)	none
Acetates (CH_3COO^-)	none
Sulfates (SO_4^{2-})	$\text{Ca}^{2+}, \text{Ba}^{2+}, \text{Pb}^{2+}$
Low Solubility Compounds	Exceptions
Carbonates (CO_3^{2-})	$\text{Li}^+, \text{Na}^+, \text{K}^+, \text{H}^+, \text{NH}_4^+$
Hydroxides (OH^-)	$\text{Li}^+, \text{Na}^+, \text{K}^+, \text{H}^+, \text{NH}_4^+, \text{Ba}^{2+}$

NOTE: From this table it can be concluded that:

all $\left[\begin{array}{l} \text{lithium} \\ \text{sodium} \\ \text{potassium} \\ \text{ammonium} \end{array} \right]$ compounds are soluble.

EXERCISE 46: PRECIPITATION REACTIONS

Write balanced equations for any of the following mixtures where a precipitate is produced. Be sure to use subscripts to indicate the form of the substance. If a reaction does not occur indicate by writing "no reaction".

1. Calcium chloride solution + sodium carbonate solution.
2. Barium hydroxide solution + dilute sulfuric acid.
3. Sodium nitrate solution + sodium chloride solution.
4. Potassium nitrate solution + lead(II) acetate solution.
5. Lead(II) nitrate solution + lead(II) acetate solution.
6. Sodium hydroxide solution + calcium chloride solution.
7. Ammonium carbonate solution + aluminium nitrate solution.
8. Magnesium nitrate solution + sodium carbonate solution.
9. Copper(II) sulfate solution + aluminium chloride solution.
10. Iron(III) sulfate solution + potassium carbonate solution.

EXERCISE 47: MISCELLANEOUS I

Write balanced equations for the following reactions. If no reaction occurs indicate by writing "no reaction". Use subscripts to indicate the phase of the substance.

1. Dilute sulfuric acid is poured over solid iron(II) hydrogencarbonate.
2. Burning of carbon in a plentiful supply of oxygen.
3. Heating of potassium carbonate.
4. Solid magnesium oxide is added to dilute nitric acid.
5. Ammonia solution is added to sodium hydroxide solution.
6. A piece of zinc is heated strongly in an atmosphere of oxygen.
7. Acetic acid solution is poured onto solid calcium oxide.
8. A copper bangle is dropped into silver nitrate solution.
9. Solid calcium hydrogencarbonate is heated with a bunsen burner.
10. Dilute sulfuric acid is poured onto solid lead(II) carbonate.
11. The "lime-water" test (carbon dioxide is bubbled through calcium hydroxide solution).
12. The "pop-test" for hydrogen (hydrogen is burnt in air).
13. The mixing of iron(II) chloride solution and sodium hydroxide solution.
14. Mercury is poured into dilute hydrochloric acid.
15. Lead(II) nitrate solution and potassium iodide solutions are mixed.
16. Neutralization of tin(II) hydroxide with dilute sulfuric acid.
17. Oxidation of calcium in air.
18. Heating of iron(II) carbonate with a bunsen burner.
19. Steam is blown over a piece of magnesium.
20. Copper(II) oxide solid is added to dilute hydrochloric acid.

PART 2: CALCULATIONS IN CHEMISTRY

CHAPTER 9: Atomic, Molecular and Formula Mass

ATOMIC MASS

In Science it is often necessary to be able to express the mass of an atom. Each atom has a mass of approximately 10^{-23}kg (0.000 000 000 000 000 001 kg). Masses of atoms in common units (e.g. in kg) are too small to be of practical value hence a relative scale has been devised.

The mass of atoms is expressed using a scale which defines the mass of one atom of the carbon - 12 (^{12}C) isotope to be exactly 12 atomic mass units.

The average value obtained on this scale for the mass of one atom of an element is termed the RELATIVE ATOMIC MASS of the element.

Relative Atomic Mass (which will be referred to simply as Atomic Mass) was previously referred to as Atomic Weight.

The (RELATIVE) ATOMIC MASS of an element is the ratio of the average mass of an atom of that element to one-twelfth of the mass of one atom of the carbon - 12 isotope.

$$\text{Atomic Mass (AM)} = \frac{\text{Average mass of one atom of the element}}{\text{One twelfth mass of one atom of carbon - 12}}$$

- NOTE: 1. Atomic Mass is a ratio of masses hence it has NO units.
2. As naturally occurring elements generally contain several isotopes, the average mass of an atom compared to one twelfth mass of a carbon - 12 atom will produce a ratio which will not be an exact whole number. However, most of the atomic masses given inside the front cover have been rounded to whole numbers.

MOLECULAR MASS, FORMULA MASS

The mass of a molecule of an element or compound can be expressed in similar terms to that used for masses of atoms. The term used is RELATIVE MOLECULAR MASS (or simply MOLECULAR MASS).

The MOLECULAR MASS is the ratio of the average mass of one molecule of a substance to one-twelfth of the mass of a carbon - 12 atom.

$$\text{Molecular Mass} = \frac{\text{Average mass of one molecule of substance}}{\text{One twelfth mass of one atom of carbon - 12}}$$

The term MOLECULAR MASS was previously referred to as molecular weight (or formula weight). It is used for all substances even though some do not exist in molecular form. The term FORMULA MASS (FM) will be used in this book to avoid this possible confusion.

DETERMINATION OF FORMULA MASS

For any substance (element or compound), the formula mass is a sum of the atomic masses of EACH atom making up the formula of the substance.

The Table of Atomic Masses found inside the cover has been used for all Type Examples and Answers to problems in this book.

TYPE EXAMPLE 1: Find the formula mass of oxygen (gas).

Formula : O_2

$$\begin{aligned}\text{FM}(\text{O}_2) &= 2 \times \text{AM}(\text{O}) \\ &= 2 \times 16 \\ &= 32\end{aligned}$$

i.e. Formula mass of oxygen (gas) is 32.

TYPE EXAMPLE 2: What is the formula mass of sulfur dioxide?

Formula : SO_2

$$\begin{aligned}\text{FM}(\text{SO}_2) &= \text{AM}(\text{S}) + 2 \times \text{AM}(\text{O}) \\ &= 32 + 2(16) \\ &= 64\end{aligned}$$

i.e. Formula mass of sulfur dioxide is 64.

SET 1: FORMULA MASS I.

Determine the formula mass of each of the following.

- | | |
|--------------------------------------|---|
| 1. Water (H_2O) | 6. Nitrogen dioxide (NO_2) |
| 2. Carbon dioxide (CO_2) | 7. Sulfuric acid (H_2SO_4) |
| 3. Nitrogen gas (N_2) | 8. Ozone (O_3) |
| 4. Chlorine gas (Cl_2) | 9. Butane (C_4H_{10}) |
| 5. Sulfur trioxide (SO_3) | 10. Ethanol ($\text{C}_2\text{H}_5\text{OH}$) |

SET 2: FORMULA MASS II.

Write down the formula and determine the formula mass of the following.

- | | |
|-----------------------|----------------------|
| 1. Hydrogen chloride | 6. Sodium sulfide |
| 2. Calcium oxide | 7. Potassium bromide |
| 3. Zinc sulfide | 8. Copper(II) oxide |
| 4. Aluminium chloride | 9. Barium chloride |
| 5. Magnesium oxide | 10. Silver sulfide |

TYPE EXAMPLE 3: Write down the formula and determine the formula mass of magnesium nitrate.

Formula : $\text{Mg}(\text{NO}_3)_2$

$$\begin{aligned}\text{FM}[\text{Mg}(\text{NO}_3)_2] &= \text{AM}(\text{Mg}) + 2 [\text{AM}(\text{N}) + 3 \times \text{AM}(\text{O})] \\ &= 24 + 2[14 + 3(16)] \\ &= 24 + 2[14 + 48] \\ &= 24 + 2(62) \\ &= 24 + 124 \\ &= 148\end{aligned}$$

i.e. Formula mass of magnesium nitrate is 148.

SET 3: FORMULA MASS III.

Write down the formula and determine the formula mass of the following.

- | | |
|--------------------------|--------------------------------|
| 1. Sodium carbonate | 6. Ammonium chloride |
| 2. Lead (II) nitrate | 7. Barium nitrate |
| 3. Magnesium hydroxide | 8. Potassium hydrogencarbonate |
| 4. Copper (II) carbonate | 9. Silver nitrate |
| 5. Iron (II) hydroxide | 10. Calcium phosphate |

SET 4: FORMULA MASS IV.

Write down the formula and determine the formula mass of the following.

- | | |
|------------------------|----------------------------|
| 1. Lead (II) hydroxide | 6. Ammonium sulfate |
| 2. Potassium sulfate | 7. Zinc hydrogencarbonate |
| 3. Ammonium sulfide | 8. Iron (II) oxide |
| 4. Aluminium carbonate | 9. Calcium hydrogensulfate |
| 5. Magnesium nitride | 10. Sodium phosphate |

TYPE EXAMPLE 4: Find the formula mass of copper (II) sulfate pentahydrate.

Formula: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

$$\begin{aligned}\text{FM}[\text{CuSO}_4 \cdot 5\text{H}_2\text{O}] &= \text{AM}(\text{Cu}) + \text{AM}(\text{S}) + 4 \times \text{AM}(\text{O}) + 5[2 \times \text{AM}(\text{H}) + \text{AM}(\text{O})] \\ &= 63.5 + 32 + 4(16) + 5[2(1) + 16] \\ &= 63.5 + 32 + 64 + 5(18) \\ &= 249.5\end{aligned}$$

i.e. Formula Mass of copper(II) sulfate pentahydrate is 249.5.

SET 5: FORMULA MASS V.

Find the formula mass of each of the following hydrated compounds.

1. Magnesium chloride hexahydrate
2. Zinc chloride monohydrate
3. Iron (II) sulfate heptahydrate
4. Sodium carbonate decahydrate
5. Ferric alum, $\text{NH}_4\text{Fe}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$

CHAPTER 11: The Mole

In Science, as previously mentioned, the mass of single atoms is far too small to be of practical value. However, masses must be used in calculations, hence masses based on Atomic (and Formula) Masses are used.

Consider the elements aluminium and carbon —

$$\begin{aligned} \frac{\text{Atomic mass of aluminium}}{\text{Atomic mass of carbon}} &= \frac{27}{12} \\ \text{i.e. } \frac{\text{the mass of 1 atom of aluminium}}{\text{the mass of 1 atom of carbon}} &= \frac{27}{12} \\ \therefore \frac{\text{the mass of 2 atoms of aluminium}}{\text{the mass of 2 atoms of carbon}} &= \frac{2 \times 27}{2 \times 12} = \frac{27}{12} \\ \text{and } \frac{\text{the mass of 10 atoms of aluminium}}{\text{the mass of 10 atoms of carbon}} &= \frac{10 \times 27}{10 \times 12} = \frac{27}{12} \\ \therefore \frac{\text{the mass of N atoms of aluminium}}{\text{the mass of N atoms of carbon}} &= \frac{N \times 27}{N \times 12} = \frac{27}{12} \end{aligned}$$

Whenever the number of aluminium and carbon atoms is the same the ratio of the mass of aluminium to the mass of carbon will be 27:12.

Therefore, if we take 27g of aluminium and 12g of carbon, each sample must contain the same number of atoms.

Similarly, if we take 1g of hydrogen or 14g of nitrogen, we must have the same number of atoms in each sample as there are atoms of carbon in a 12g sample of carbon.

That is, whenever we take a sample of an element which has a mass equal to its Atomic Mass expressed in grams we have the same number of atoms in that sample as there are atoms of carbon in a 12g sample of carbon.

The very large number of atoms contained in such a sample is called Avogadro's Number.

$$\text{Avogadro's Number} = 6.02 \times 10^{23} \text{ (approximately)}$$

Similarly it can be shown that Avogadro's number of molecules of a substance has a mass which is equal to its Formula Mass expressed in grams.

Avogadro's Number (6.02×10^{23}) of particles of a substance is known as a MOLE of that substance.

DEFINITION

There are two acceptable definitions of the MOLE —

1. A MOLE of a substance is that quantity of that substance which contains AVOGADRO'S NUMBER (6.02×10^{23}) of constituent particles.
2. A MOLE of a substance is that quantity of that substance which has a mass equal to its atomic (or formula) mass expressed in grams.

FROM DEFINITION 1 OF THE MOLE

1 mole of a substance contains 6.02×10^{23} constituent particles.

$\therefore$ n moles of a substance contains $n \times 6.02 \times 10^{23}$ constituent particles.

i.e. number of particles = number of moles $\times$ (6.02×10^{23})

Rearranging this equation gives,

$$\text{number of moles, } n = \frac{\text{number of particles}}{6.02 \times 10^{23}}$$

TYPE EXAMPLE 11: Calculate the number of moles of ammonia molecules in 60.2×10^{23} molecules of ammonia.

METHOD A (PROPORTION)

1 mole NH_3 contains 6.02×10^{23} molecules
 x moles NH_3 contains 60.2×10^{23} molecules

$$\frac{x}{1} = \frac{60.2 \times 10^{23}}{6.02 \times 10^{23}}$$

$$\therefore x = 10 \text{ moles}$$

i.e. 60.2×10^{23} molecules NH_3 contains 10 moles NH_3 .

METHOD B (FORMULA)

$$\begin{aligned} \text{number of moles of } \text{NH}_3 &= n(\text{NH}_3) \\ &= ? \end{aligned} \qquad n(\text{NH}_3) = \frac{\text{number molecules } \text{NH}_3}{6.02 \times 10^{23}}$$

$$\begin{aligned} \text{number molecules } \text{NH}_3 &= 60.2 \times 10^{23} \\ &= \frac{60.2 \times 10^{23}}{6.02 \times 10^{23}} \\ &= 10 \text{ moles} \end{aligned}$$

i.e. 60.2×10^{23} molecules NH_3 contains 10 moles NH_3 .

SET 13: NUMBER MOLECULES — MOLES

Calculate the number of moles of

1. water molecules in 602×10^{23} molecules water
2. silver atoms in 0.602×10^{23} atoms silver
3. carbon dioxide in 3.01×10^{23} molecules carbon dioxide
4. hydrogen molecules in 120.4×10^{23} molecules hydrogen
5. sulfuric acid in 1.806×10^{23} molecules sulfuric acid

TYPE EXAMPLE 12: Determine the number of molecules of oxygen in 0.4 mole oxygen (O_2).

METHOD A (PROPORTION)

1 mole O_2 contains 6.02×10^{23} molecules
0.4 mole O_2 contains x molecules

$$\begin{aligned}\frac{0.4}{1} &= \frac{x}{6.02 \times 10^{23}} \\ \therefore x &= 0.4 \times 6.02 \times 10^{23} \\ &= 2.408 \times 10^{23} \text{ molecules}\end{aligned}$$

i.e. 0.4 mole O_2 contains 2.408×10^{23} molecules O_2 .

METHOD B (FORMULA)

$$n(O_2) = 0.4 \text{ mole}$$

number molecules $O_2 = ?$

$$\begin{aligned}n(O_2) &= \frac{\text{number molecules } O_2}{6.02 \times 10^{23}} \\ 0.4 &= \frac{\text{number molecules } O_2}{6.02 \times 10^{23}}\end{aligned}$$

$$\begin{aligned}\therefore \text{number molecules } O_2 &= 0.4 \times 6.02 \times 10^{23} \\ &= 2.408 \times 10^{23}\end{aligned}$$

i.e. 0.4 mole O_2 contains 2.408×10^{23} molecules O_2 .

SET 13: (continued)

Determine the number of

6. atoms of copper in 0.2 mole copper.
7. molecules of nitrogen dioxide in 5 moles nitrogen dioxide.
8. atoms of carbon in 2 moles carbon.
9. molecules of chlorine in 0.1 mole chlorine (Cl_2).
10. molecules of benzene in 2.5 moles benzene.

FROM DEFINITION 2 OF THE MOLE, to determine the mass of 1 mole of ANY substance simply express its formula mass in grams.

TYPE EXAMPLE 13: Determine the mass of 1 mole of —

- (a) hydrogen
- (b) chlorine atoms
- (c) calcium oxide

(a) Hydrogen, H_2 :

$$\begin{aligned}FM[H_2] &= 2 \times AM(H) \\ &= 2 \times 1 \\ &= 2\end{aligned}$$

$\therefore$ Mass of 1 mole of hydrogen = 2g.

(b) Chlorine atoms, Cl:

$$AM[Cl] = 35.5$$

$\therefore$ Mass of 1 mole chlorine atoms = 35.5g.

(c) Calcium oxide, CaO:

$$\begin{aligned}FM[CaO] &= AM(Ca) + AM(O) \\ &= 40 + 16 \\ &= 56\end{aligned}$$

$\therefore$ Mass of 1 mole calcium oxide = 56g.

SET 14: MOLE — MASS I

What is the mass of 1 mole of each of the following?

1. Nitrogen (gas)
2. Iron
3. Oxygen atoms
4. Silver bromide
5. Sodium hydroxide
6. Lead (II) oxide
7. Magnesium hydrogensulfate
8. Aluminium hydroxide
9. Zinc carbonate
10. Barium hydrogencarbonate

TYPE EXAMPLE 14: Determine the mass of 0.4 mole magnesium carbonate.

METHOD A (PROPORTION)

Formula: $MgCO_3$

$$\begin{aligned}FM[MgCO_3] &= 24 + 12 + 3(16) \\ &= 84\end{aligned}$$

1 mole magnesium carbonate has mass of 84g

0.4 mole magnesium carbonate has mass of x g

$$\frac{x}{84} = \frac{0.4}{1}$$

$$\therefore x = 84 \times 0.4$$

$$= 33.6\text{g}$$

i.e. 0.4 mole MgCO_3 has a mass of 33.6g.

METHOD B (FORMULA)

A formula can be derived by simple proportion.

The mass of 1 mole of ANY substance is known as the MOLAR MASS (M) of that substance and is the formula mass expressed in grams.

If the mass of 1 mole = M,
then the mass (m) of n moles = n × mass of 1 mole
= n × M

$$\text{i.e. } m = n \times M$$

This formula is more commonly referred to in the rearranged form—

$$n = \frac{m}{M}$$

where n = number of moles
m = mass of substance (g)
M = molar mass (g)

Note: Strictly speaking the units of M should be g.mole^{-1}

Applying this formula to TYPE EXAMPLE 14.

METHOD B (FORMULA)

Formula: MgCO_3

$$\text{FM}[\text{MgCO}_3] = 24 + 12 + 48$$

$$= 84$$

$$n(\text{MgCO}_3) = 0.4 \text{ mole}$$

$$m(\text{MgCO}_3) = ?$$

$$M(\text{MgCO}_3) = 84\text{g}$$

$$n = \frac{m}{M}$$

$$0.4 = \frac{m}{84}$$

$$\therefore m = 0.4 \times 84$$

$$= 33.6\text{g}$$

Again, mass of 0.4 mole MgCO_3 is 33.6g.

SET 15: MOLE — MASS II

Determine the mass of —

1. 2 moles aluminium
2. 5 moles oxygen gas
3. 0.2 mole zinc oxide
4. 1.5 moles barium carbonate
5. 0.6 mole copper (II) chloride
6. 0.1 mole iron (II) nitrate
7. 3.2 moles potassium sulfate
8. 2.5 moles sodium hydrogencarbonate
9. 0.5 mole mercury (II) nitrate
10. 0.4 mole iron (II) sulfide

TYPE EXAMPLE 15: Calculate how many moles of water are in 72g water.

Formula: H_2O

$$\text{FM}[\text{H}_2\text{O}] = 2(1) + 16 = 18$$

Mass of 1 mole water, $M(\text{H}_2\text{O}) = 18\text{g}$

METHOD A (PROPORTION)

1 mole H_2O has a mass of 18g

x mole H_2O has a mass of 72g

$$\frac{x}{1} = \frac{72}{18}$$

$$\therefore x = 4 \text{ moles}$$

i.e. 72g H_2O contains 4 moles H_2O

METHOD B (FORMULA)

$$M(\text{H}_2\text{O}) = 18\text{g}$$

$$m(\text{H}_2\text{O}) = 72\text{g}$$

$$n(\text{H}_2\text{O}) = ?$$

$$n = \frac{m}{M}$$

$$= \frac{72}{18}$$

$$= 4 \text{ moles}$$

i.e. 72g H_2O contains 4 moles H_2O .

SET 16: MASS — MOLE

Determine the number of moles of

1. magnesium in 48g magnesium
2. hydrogen in 1g hydrogen gas
3. carbon dioxide in 4.4g carbon dioxide
4. calcium sulfate in 27.2g calcium sulfate
5. potassium hydroxide in 22.4g potassium hydroxide
6. zinc nitrate in 947g zinc nitrate
7. ammonium carbonate in 288g ammonium carbonate
8. lead (II) chloride in 55.6g lead (II) chloride
9. sodium sulfate in 85.2g sodium sulfate
10. silver chloride in 114.8g silver chloride

TYPE EXAMPLE 16: If 1.5 moles of benzene has a mass of 117g, determine the formula mass of benzene.

METHOD A (PROPORTION)

1.5 moles has mass of 117g.
1 mole benzene has mass of xg

$$\begin{aligned}\frac{x}{117} &= \frac{1}{1.5} \\ \therefore x &= \frac{117}{1.5} \\ &= 78\text{g}\end{aligned}$$

From the definition of a mole, the mass of 1 mole of any substance is its formula mass expressed in grams.

$\therefore$ Formula mass of benzene is 78

METHOD B (FORMULA)

n (benzene) = 1.5 moles

m (benzene) = 117g

M (benzene) = ?

$$\begin{aligned}n &= \frac{m}{M} \\ 1.5 &= \frac{117}{M} \\ \therefore M &= \frac{117}{1.5} \\ &= 78\text{g}\end{aligned}$$

i.e. Formula mass of benzene is 78.

SET 17: FORMULA MASS

Determine the formula mass of the substance in each of the questions below, if

1. 3 moles acetylene has a mass of 78g
2. 2 moles phenol has a mass of 188g
3. 0.1 mole calcium hypochlorite has a mass of 14.3g
4. 0.2 mole methanol has a mass of 6.4g
5. 1.5 moles ethylene has a mass of 42g
6. the mass of 5 moles propane is 220g
7. the mass of 0.4 mole ethanol is 18.4g
8. the mass of 2.5 moles of a gas is 115g
9. the mass of 0.1 mole ethane is 3g
10. the mass of 0.3 mole of a silvery metal is 15.6g.

TYPE EXAMPLE 17: Determine the number of moles of atoms of hydrogen, sulfur and oxygen in 1 mole sulfuric acid.

Formula: H_2SO_4

Examination of the formula shows 1 molecule of H_2SO_4 contains—

- (a) 2 atoms H
- (b) 1 atom S
- (c) 4 atoms O

$\therefore$ Avogadro's Number of molecules of H_2SO_4 must contain —

- (a) $2 \times$ Avogadro's Number of H atoms.
- (b) $1 \times$ Avogadro's Number of S atoms.
- (c) $4 \times$ Avogadro's Number of O atoms.

i.e. 1 mole H_2SO_4 contains

- (a) 2 moles H atoms
- (b) 1 mole S atoms
- (c) 4 moles O atoms

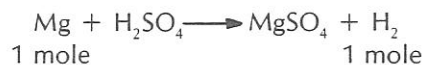
TYPE EXAMPLE 18: Find the number of moles of potassium ions in 0.2 mole potassium sulfate.

Formula: K_2SO_4

MASS-MASS DETERMINATIONS:

TYPE EXAMPLE 32: What mass of hydrogen is produced when 6g magnesium is added to excess sulfuric acid?

METHOD A (PROPORTION)



$$\text{FM}(\text{Mg}) = 24$$

$$\text{FM}(\text{H}_2) = 2$$

1 mole magnesium produces 1 mole hydrogen (from equation)

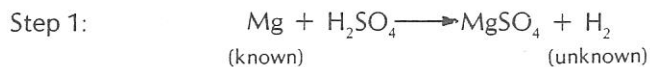
$\therefore$ 24g magnesium produces 2g hydrogen

6g magnesium produces xg hydrogen

$$\begin{aligned} \frac{x}{2} &= \frac{6}{24} \\ \therefore x &= \frac{2 \times 6}{24} \\ &= 0.5\text{g} \end{aligned}$$

i.e. 6g Mg produces 0.5g H_2 .

METHOD B (FORMULA)



Step 2:

$$m(\text{Mg}) = 6\text{g}$$

$$n(\text{Mg}) = ?$$

$$M(\text{Mg}) = 24\text{g}$$

$$n = \frac{m}{M}$$

$$= \frac{6}{24}$$

$$= 0.25 \text{ mole}$$

$$\text{FM}(\text{Mg}) = 24$$

$$\text{FM}(\text{H}_2) = 2$$

Step 3:

$$\frac{n(\text{H}_2)}{n(\text{Mg})} = \frac{1}{1}$$

Step 4:

$$n(\text{Mg}) = 0.25 \text{ mole}$$

$$n(\text{H}_2) = ?$$

$$\frac{n(\text{H}_2)}{0.25} = \frac{1}{1}$$

$$\therefore n(\text{H}_2) = 0.25 \text{ mole}$$

Step 5:

$$n(\text{H}_2) = 0.25 \text{ mole}$$

$$m(\text{H}_2) = ?$$

$$M(\text{H}_2) = 2\text{g}$$

$$n = \frac{m}{M}$$

$$0.25 = \frac{m}{2}$$

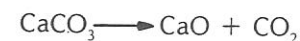
$$\therefore m = 0.25 \times 2$$

$$= 0.5\text{g}$$

i.e. 6g Mg produces 0.5g H_2 .

SET 32: MASS-MASS DETERMINATIONS I

1. 25g calcium carbonate is strongly heated.



Determine the mass of calcium oxide produced.

2. In an experiment 3.6g water is electrolysed.



Find the mass of

(a) hydrogen, and

(b) oxygen produced.

3. 20g sodium hydroxide is neutralized by dilute hydrochloric acid.

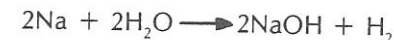


How many grams of sodium chloride can be recovered?

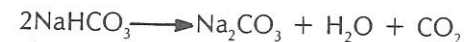
4. Determine the mass of calcium sulfate produced when 5.6g calcium oxide is neutralized by sulfuric acid.



5. 6.9g sodium metal is reacted with water. Find the mass of sodium hydroxide produced.



6. 420g sodium hydrogencarbonate is strongly heated.



Determine the mass of —

(a) sodium carbonate left, and

(b) carbon dioxide produced.

7. Determine the mass of hydrochloric acid required to completely react with 16.8g magnesium carbonate.



8. 6.35g copper is dissolved when placed in a solution of silver nitrate.



How many grams of

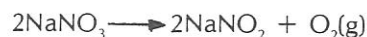
- (a) silver is deposited, and
(b) copper(II) nitrate can be recovered?

9. Sufficient potassium carbonate solution is added to 34.1g zinc chloride to completely precipitate the zinc as zinc carbonate.



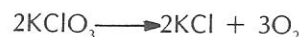
What mass of zinc carbonate is deposited?

10. Determine the mass of
(a) sodium nitrite left, and
(b) oxygen gas evolved
when 17g sodium nitrate is strongly heated

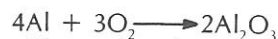


SET 33: MASS-MASS DETERMINATIONS II

1. Determine the mass of potassium chlorate which must be decomposed by heating to produce 9.6g oxygen.

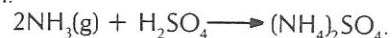


2. 10.8g aluminium powder is burnt in air.



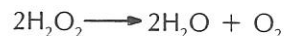
Find the mass of aluminium oxide produced.

3. 66g ammonium sulfate is produced when ammonia gas is bubbled through a sulfuric acid solution.



What mass of ammonia is required?

4. Calculate the mass of hydrogen peroxide which upon catalytic decomposition will produce 4g oxygen.



5. Determine the mass of sodium carbonate which must be added to a calcium nitrate solution to precipitate all of the calcium as 8g of calcium carbonate.



6. Find the mass of chlorine produced when 4.35g manganese dioxide is reacted with concentrated hydrochloric acid.



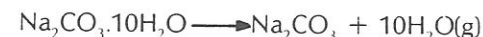
7. What mass of barium chloride is produced in the reaction between hydrochloric acid and barium carbonate which produces 8.8g carbon dioxide?



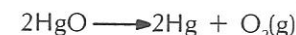
8. Determine the mass of carbon dioxide formed when 16kg methane (CH_4) burns in air.



9. 71.5g sodium carbonate decahydrate is heated until it is completely dehydrated. Calculate the mass of water (as steam) driven off.



10. A sample of mercury(II) oxide on heating produced 1.6g oxygen gas.



Determine

- (a) the mass of mercury(II) oxide used, and
(b) the mass of mercury remaining.

MASS-VOLUME DETERMINATIONS:

In Chapter II the term MOLAR VOLUME was defined as the volume occupied by 1 mole of a gas.

It was shown that the Molar Volume for any gas is 22.4L when measured under S.T.P. conditions.

The formula for mole — volume determinations was derived:

$$n = \frac{V}{\text{Molar Volume}} \quad (\text{any conditions})$$

OR

$$n = \frac{V}{22.4}$$

Where

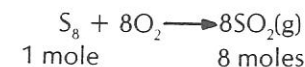
n = moles of gas

V = volume of gas in litres

22.4L = Molar Volume (at S.T.P.)

TYPE EXAMPLE 33: What volume of sulfur dioxide is produced (at S.T.P.) by burning 3.2g sulfur?

METHOD A (PROPORTION)



$$\text{FM}(\text{S}_8) = 8(32) = 256$$

ANSWERS — PART 2

SET 1

- | | | | | |
|-------|-------|-------|-------|--------|
| 1. 18 | 2. 44 | 3. 28 | 4. 71 | 5. 80 |
| 6. 46 | 7. 98 | 8. 48 | 9. 58 | 10. 46 |

SET 2

- | | | |
|------------------------------|--------------|----------------------------|
| 1. HCl, 36.5 | 2. CaO, 56 | 3. ZnS, 97.4 |
| 4. AlCl ₃ , 133.5 | 5. MgO, 40 | 6. Na ₂ S, 78 |
| 7. KBr, 119 | 8. CuO, 79.5 | 9. BaCl ₂ , 208 |
| 10. Ag ₂ S, 248 | | |

SET 3

- | | | |
|---|--|-----------------------------|
| 1. Na ₂ CO ₃ , 106 | 2. Pb(NO ₃) ₂ , 331 | 3. Mg(OH) ₂ , 58 |
| 4. CuCO ₃ , 123.5 | 5. Fe(OH) ₂ , 90 | 6. NH ₄ Cl, 53.5 |
| 7. Ba(NO ₃) ₂ , 261 | 8. KHCO ₃ , 100 | 9. AgNO ₃ , 170 |
| 10. Ca ₃ (PO ₄) ₂ , 310 | | |

SET 4

- | | | |
|--|---|--|
| 1. Pb(OH) ₂ , 241 | 2. K ₂ SO ₄ , 174 | 3. (NH ₄) ₂ S, 68 |
| 4. Al ₂ (CO ₃) ₃ , 234 | 5. Mg ₃ N ₂ , 100 | 6. (NH ₄) ₂ SO ₄ , 132 |
| 7. Zn(HCO ₃) ₂ , 187.4 | 8. FeO, 72 | 9. Ca(HSO ₄) ₂ , 234 |
| 10. Na ₃ PO ₄ , 164 | | |

SET 5

- | | | | | |
|--------|----------|--------|--------|--------|
| 1. 203 | 2. 154.4 | 3. 278 | 4. 286 | 5. 482 |
|--------|----------|--------|--------|--------|

SET 6

- | | |
|------------------------------|-----------------------|
| 1. C 27.3%; O 72.7% | 2. N 82.4%; H 17.6% |
| 3. Ca 40%; C 12%; O 48% | 4. Mg 25.3%; Cl 74.7% |
| 5. Ag 63.5%; N 8.2%; O 28.2% | |

SET 7

- | | | | | |
|----------|----------|----------|----------|-----------|
| 1. 11.1% | 2. 40% | 3. 75% | 4. 57.1% | 5. 32.4% |
| 6. 18.9% | 7. 74.5% | 8. 18.7% | 9. 92.6% | 10. 23.5% |

SET 8

- | | | | | |
|----------|----------|----------|----------|----------|
| 1. 72.9% | 2. 22.5% | 3. 47.7% | 4. 84.2% | 5. 83.6% |
|----------|----------|----------|----------|----------|

SET 9

- | | | | | |
|----------|----------|----------|----------|----------|
| 1. 14.8% | 2. 51.2% | 3. 21.1% | 4. 39.9% | 5. 22.4% |
|----------|----------|----------|----------|----------|

SET 10

- | | | | | |
|----------|---------|---------|---------|-----------|
| 1. 4g | 2. 6.4g | 3. 2.8g | 4. 254g | 5. 130.8g |
| 6. 68.5g | 7. 1.2g | 8. 6.9g | 9. 1g | 10. 8.4g |

SET 11

- | | | | | |
|----------|-----------|----------|----------|----------|
| 1. 14g | 2. 117g | 3. 11.6g | 4. 51g | 5. 26.4g |
| 6. 46.1g | 7. 116.5g | 8. 44.7g | 9. 35.6g | 10. 8.7g |

SET 12

- | | |
|---|--|
| 1(a) CaCl ₂ , 111 | (b) NH ₄ NO ₃ , 80 |
| 2(a) Ba(HSO ₄) ₂ , 331 | (b) Zn(OH) ₂ , 99.4 |
| 3. Mg, 20%; S, 26.7%; O, 53.3% | 4. 8.7% |
| 5. 62.9% | 6. 6.4g |
| 8. 39g | 9. 89g |
| | 10. 54.25g |

SET 13

- | | | | | |
|---------------------------|--------|----------------------------|---------------------------|--------|
| 1. 100 | 2. 0.1 | 3. 0.5 | 4. 20 | 5. 0.3 |
| 6. 1.204×10^{23} | | 7. 30.1×10^{23} | 8. 12.04×10^{23} | |
| 9. 0.602×10^{23} | | 10. 15.05×10^{23} | | |

SET 14

- | | | | | |
|---------|---------|--------|-----------|----------|
| 1. 28g | 2. 56g | 3. 16g | 4. 188g | 5. 40g |
| 6. 223g | 7. 218g | 8. 78g | 9. 125.4g | 10. 259g |

SET 15

- | | | | | |
|--------|-----------|-----------|-----------|-----------|
| 1. 54g | 2. 160g | 3. 16.28g | 4. 295.5g | 5. 80.7g |
| 6. 18g | 7. 556.8g | 8. 210g | 9. 162.5g | 10. 35.2g |

SET 16

- | | | | | |
|------|--------|--------|--------|---------|
| 1. 2 | 2. 0.5 | 3. 0.1 | 4. 0.2 | 5. 0.4 |
| 6. 5 | 7. 3 | 8. 0.2 | 9. 0.6 | 10. 0.8 |

SET 17

- | | | | | |
|-------|-------|--------|-------|--------|
| 1. 26 | 2. 94 | 3. 143 | 4. 32 | 5. 28 |
| 6. 44 | 7. 46 | 8. 46 | 9. 30 | 10. 52 |

SET 18

- | | | |
|--|-----------------------------------|---------------------------|
| 1. 0.2 | 2. H, 2; N, 2; O, 6 | 3. Cu, 2.5; S, 2.5; O, 10 |
| 4. Ca ²⁺ , 0.4; OH ⁻ , 0.8 | 5. 0.602×10^{23} of each | |
| 6. H, 36.12×10^{23} ; S, 18.06×10^{23} | | |
| 7(a) N, 0.2; H, 0.6 (b) NH ₃ , 1.204×10^{23} ; N, 1.204×10^{23} ; H, 3.612×10^{23} | | |
| 8. N, 1; H, 4; C, 0.5; O, 1.5 | | |

SET 19

- | | | | | |
|--------|---------|---------|--------|---------|
| 1. 0.1 | 2. 0.4 | 3. 2 | 4. 1.5 | 5. 0.4 |
| 6. 3 | 7. 0.15 | 8. 0.75 | 9. 0.4 | 10. 1.4 |

SET 20

- | | | | | |
|--------|----------|----------|----------|----------|
| 1. 32g | 2. 4g | 3. 14g | 4. 6.4g | 5. 22.4g |
| 6. 48g | 7. 11.7g | 8. 2010g | 9. 12.8g | 10. 48g |

SET 21

- | | |
|-------------------|---|
| 1.(a) 40g (b) 73g | 2.(a) 0.602×10^{23} (b) 36.12×10^{23} |
| 3.(a) 0.2 (b) 0.4 | 4. 0.5 |
| 6. 249.5g | 7.(a) 46.4g (b) 390g |
| 9.(a) 0.1 (b) 1 | 10. 60g |
| 12. 70 | 13.(a) 255g (b) 53g |
| 15. 5.36 | 16. 0.4 |
| 18. 16.8g | 19. 0.4 |

SET 22

- | | | | | |
|----------------------------|---|------------------------------------|-----------------------------|-----------------------------|
| 1. 0.2 mol L ⁻¹ | 2. 165.5 g L ⁻¹ | 3. 50 g L ⁻¹ | 1.25 mol L ⁻¹ | 4. 0.25 mol L ⁻¹ |
| 21.25 g L ⁻¹ | 5. 0.279 g L ⁻¹ | 2.56 g L ⁻¹ | 6. 6.15 mol L ⁻¹ | 0.0216 mol L ⁻¹ |
| 7.4.56 mol L ⁻¹ | 7.61 mol L ⁻¹ | 8. 0.0341 mol L ⁻¹ | 0.0204 mol L ⁻¹ | 9. 0.0384 g L ⁻¹ |
| 0.0736 g L ⁻¹ | 10. (a) about 53° (b) (i) about 525 g L ⁻¹ (ii) about 310g L ⁻¹ | (d) about 0.88 mol L ⁻¹ | (e) about 32°C | |

SET 23

- | | | | | |
|-----------|---------|-----------|---------|---------------------|
| 1. 156.8L | 2. 56L | 3. 0.896L | 4. 224L | 5. 0.0224L (22.4mL) |
| 6. 0.1 | 7. 0.05 | 8. 8 | 9. 70 | 10. 30 |

SET 24

- | | | | | |
|----------|----------|----------|-----------|----------|
| 1. 44.8L | 2. 5.6L | 3. 8.96L | 4. 89.6L | 5. 1.12L |
| 6. 4.48L | 7. 33.6L | 8. 2.8L | 9. 24.64L | 10. 56L |

SET 25

- | | | | | |
|--------|------------|----------|----------|----------|
| 1. 11g | 2. 3.55g | 3. 0.6g | 4. 160g | 5. 10g |
| 6. 40g | 7. 127.75g | 8. 2.55g | 9. 34.5g | 10. 3.2g |

SET 26

- | | | | | |
|-------------------|--------|-----------|---------|-----------|
| 1. 53.76L | 2. 2.5 | 3. 4.48L | 4. 1.4L | 5. 38.4g |
| 6.(a)7.5 (b) 330g | 7. 0.4 | 8. 15.68L | 9. 0.75 | 10. 45.6g |

SET 27 qualitative, no answers**SET 28**

- | | | | | |
|------|-------------------|---------|-------------------|--------------------|
| 1. 4 | 2.(a) 0.4 (b) 0.2 | 3. 0.2 | 4.(a) 0.3 (b) 0.3 | 5.(a) 1.5 (b) 0.75 |
| 6. 1 | 7. 0.22 | 8. 0.02 | 9. 0.2 | 10. 0.4 |

SET 29

- | | | |
|----------------------|---------------------|-----------|
| 1. 102.4g | 2.(a) 5.3g (b) 2.2g | 3. 17.75g |
| 4.(a) 14.9g (b) 9.6g | 5. 0.72g | |

SET 30

- | | | |
|-----------------------|---------------------|---------|
| 1. 4.8g | 2.(a) 68g (b) 196g | 3. 8.7g |
| 4.(a) 6.175g (b) 2.2g | 5.(a) 31.2g (b) 0.8 | |

SET 31

- | | | |
|----------|---------|---------------------|
| 1. 0.2 | 2. 0.5 | 3.(a) 0.1 (b) 0.05 |
| 4. 0.4 | 5. 0.25 | 6.(a) 0.25 (b) 0.25 |
| 7. 0.5 | 8. 8 | 9. 0.4 |
| 10. 0.08 | | |

SET 32

- | | | |
|-----------------------|------------------------|---------------------|
| 1. 14g | 2.(a) 0.4g (b) 3.2g | 3. 29.25g |
| 4. 13.6g | 5. 12g | 6.(a) 265g (b) 110g |
| 7. 14.6g | 8.(a) 21.6g (b) 18.75g | 9. 31.35g |
| 10.(a) 13.8g (b) 3.2g | | |

SET 33

- | | | | | |
|------------------------|----------|---------|---------|----------|
| 1. 24.5g | 2. 20.4g | 3. 17g | 4. 8.5g | 5. 8.48g |
| 6. 3.55g | 7. 41.6g | 8. 44kg | 9. 45g | |
| 10.(a) 21.7g (b) 20.1g | | | | |

SET 34

- | | | |
|---------------------|-----------------------|----------------------|
| 1. 4.48L | 2. 1.12L | 3. 0.224L |
| 4.(a) 5.6L (b) 2.8L | 5. 2.8L | 6.(a) 1.12L (b) 6.9g |
| 7. 3.36L | 8.(a) 2.24L (b) 6.72L | 9.(a) 336L (b) 1375g |
| 10. 2.24L | | |

SET 35

- | | | |
|--------------------------|----------|------------------|
| 1. 0.2 | 2. 18g | 3. 14g |
| 4.(a) 10.7g (b) 7.4g | 5. 7.1kg | 6.(a) 4 (b) 400g |
| 7. 9.6g | 8. 15.6g | 9. 64.35g |
| 10.(a) 3.175g (b) 9.375g | | |